

Titanic Exploratory Data Analysis (EDA)

This notebook performs exploratory data analysis (EDA) on the Titanic dataset (`train.csv` , `test.csv` , `gender_submission.csv`).

- Author: Generated by ChatGPT
- Tools: Python (Pandas, Matplotlib, Seaborn)

```
In [2]: # 1. Import Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

pd.set_option('display.max_columns', None)
```

```
In [4]: # 2. Load Data
train_df = pd.read_csv('train.csv')
test_df = pd.read_csv('test.csv')
gender_df = pd.read_csv('gender_submission.csv')

print("Train shape:", train_df.shape)
print("Test shape:", test_df.shape)
print("Gender_submission shape:", gender_df.shape)

train_df.head()
```

Train shape: (891, 12)

Test shape: (418, 11)

Gender_submission shape: (418, 2)

Out[4]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

In [5]:

```
# 3. Basic Info & Summary
train_df.info()
train_df.describe()

print("Missing values:\n", train_df.isnull().sum())
print("Survived:\n", train_df['Survived'].value_counts())
print("Pclass:\n", train_df['Pclass'].value_counts())
print("Sex:\n", train_df['Sex'].value_counts())
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column      Non-Null Count  Dtype
---  -
 0   PassengerId 891 non-null    int64
 1   Survived    891 non-null    int64
 2   Pclass      891 non-null    int64
 3   Name        891 non-null    object
 4   Sex         891 non-null    object
 5   Age         714 non-null    float64
 6   SibSp       891 non-null    int64
 7   Parch       891 non-null    int64
 8   Ticket      891 non-null    object
 9   Fare        891 non-null    float64
10   Cabin       204 non-null    object
11   Embarked    889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
Missing values:
  PassengerId      0
Survived          0
Pclass            0
Name              0
Sex               0
Age              177
SibSp             0
Parch             0
Ticket            0
Fare              0
Cabin            687
Embarked         2
dtype: int64
Survived:
Survived
0    549
1    342
Name: count, dtype: int64
Pclass:
Pclass
3    491

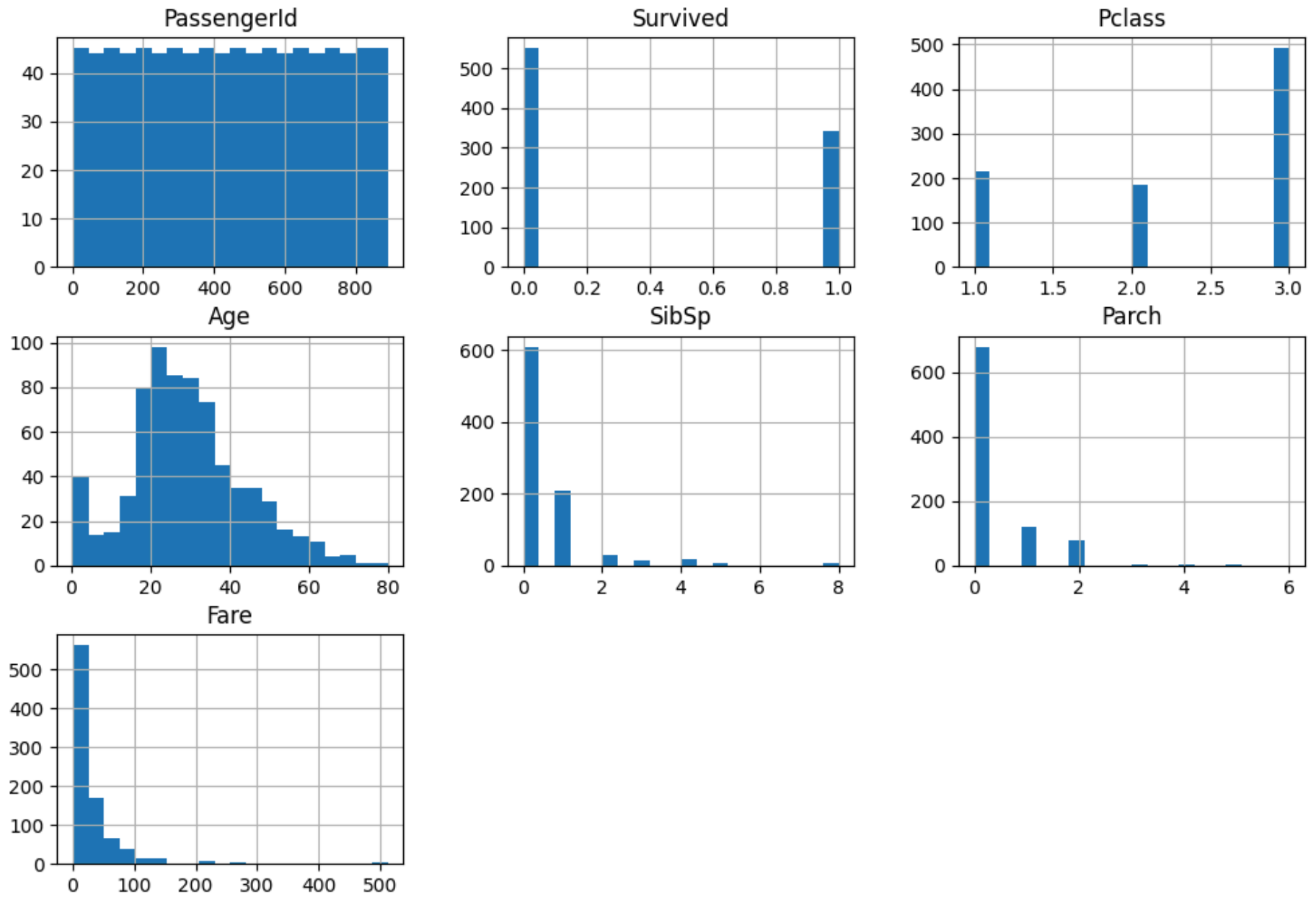
```

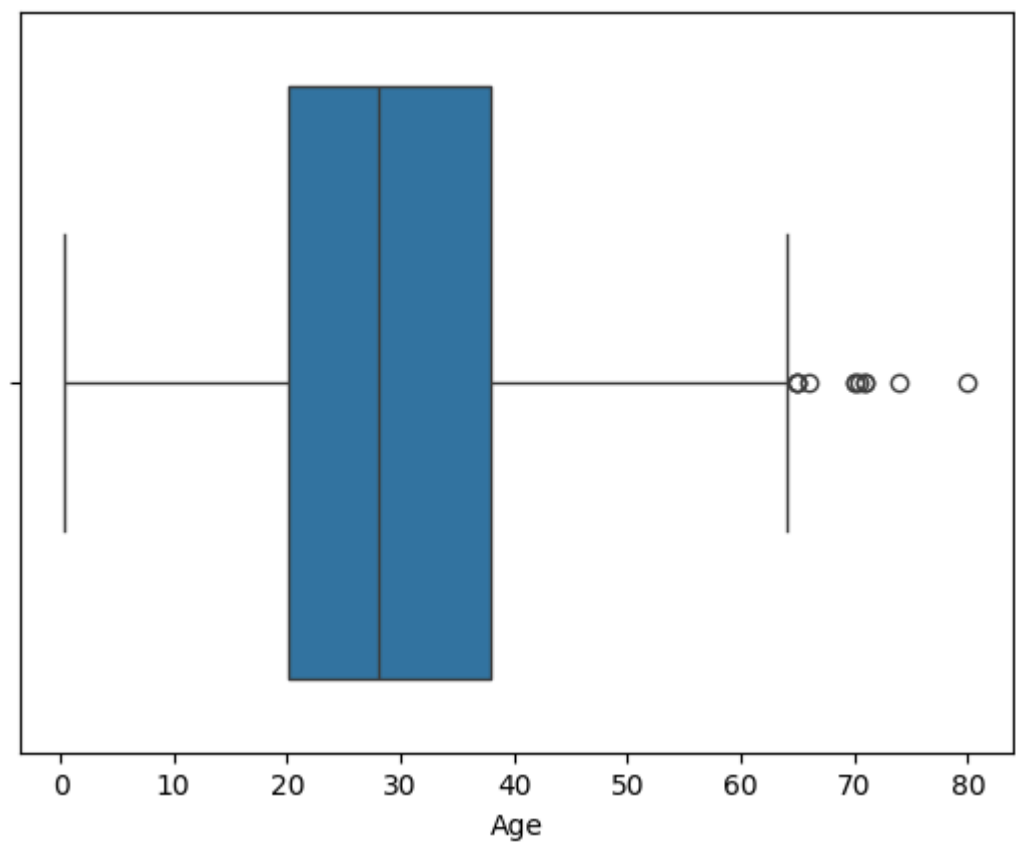
```
1    216
2    184
Name: count, dtype: int64
Sex:
  Sex
male    577
female  314
Name: count, dtype: int64
```

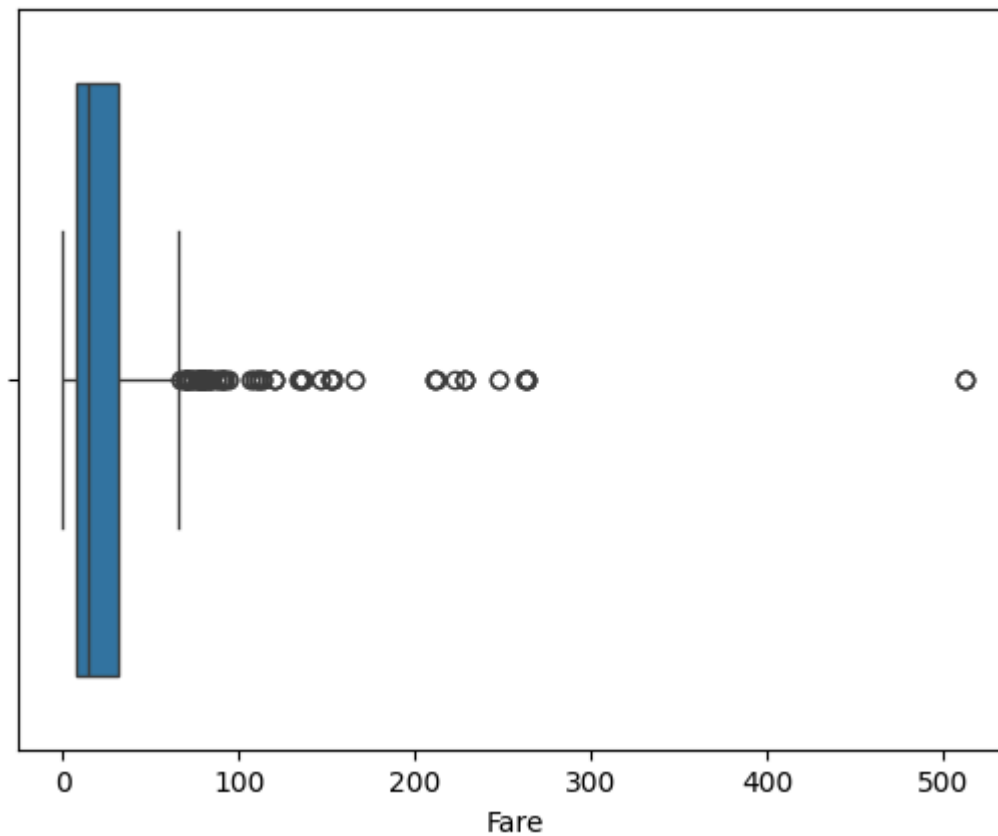
```
In [6]: # 4. Univariate Analysis
train_df.hist(figsize=(12,8), bins=20)
plt.show()

sns.boxplot(x=train_df['Age'])
plt.show()

sns.boxplot(x=train_df['Fare'])
plt.show()
```



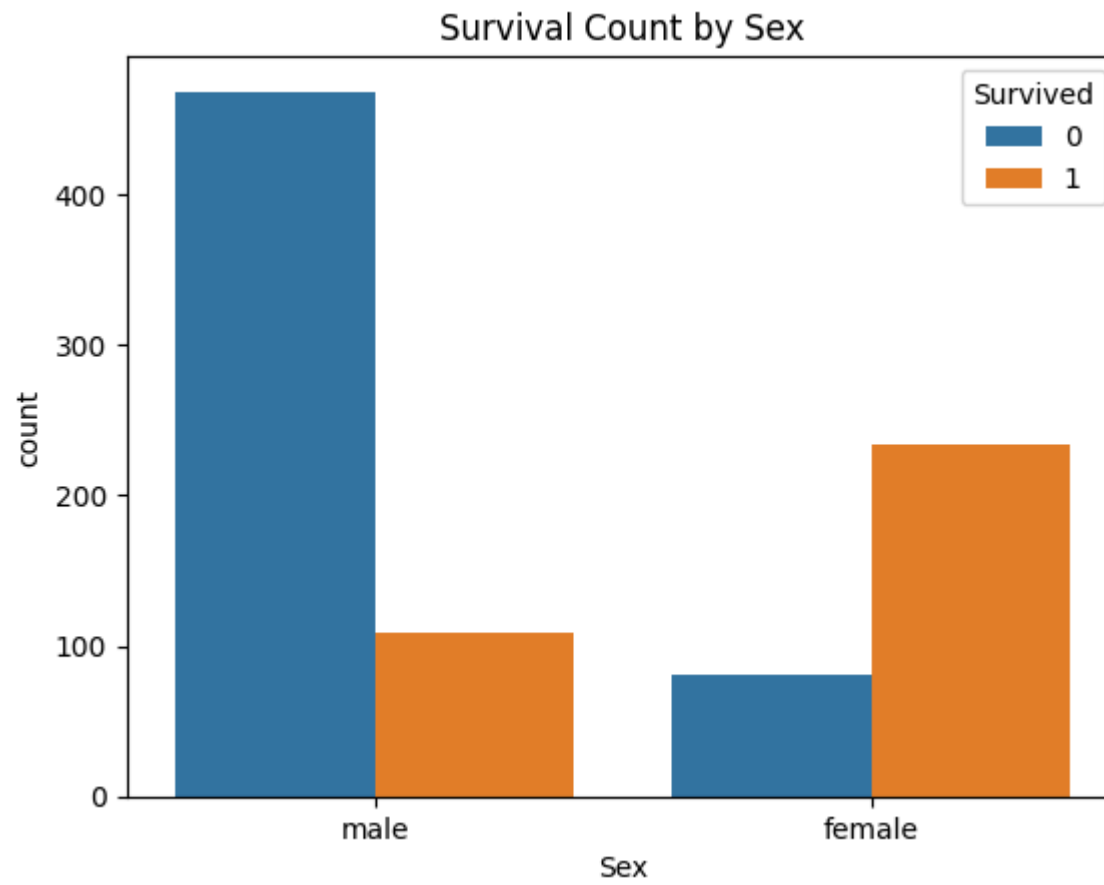


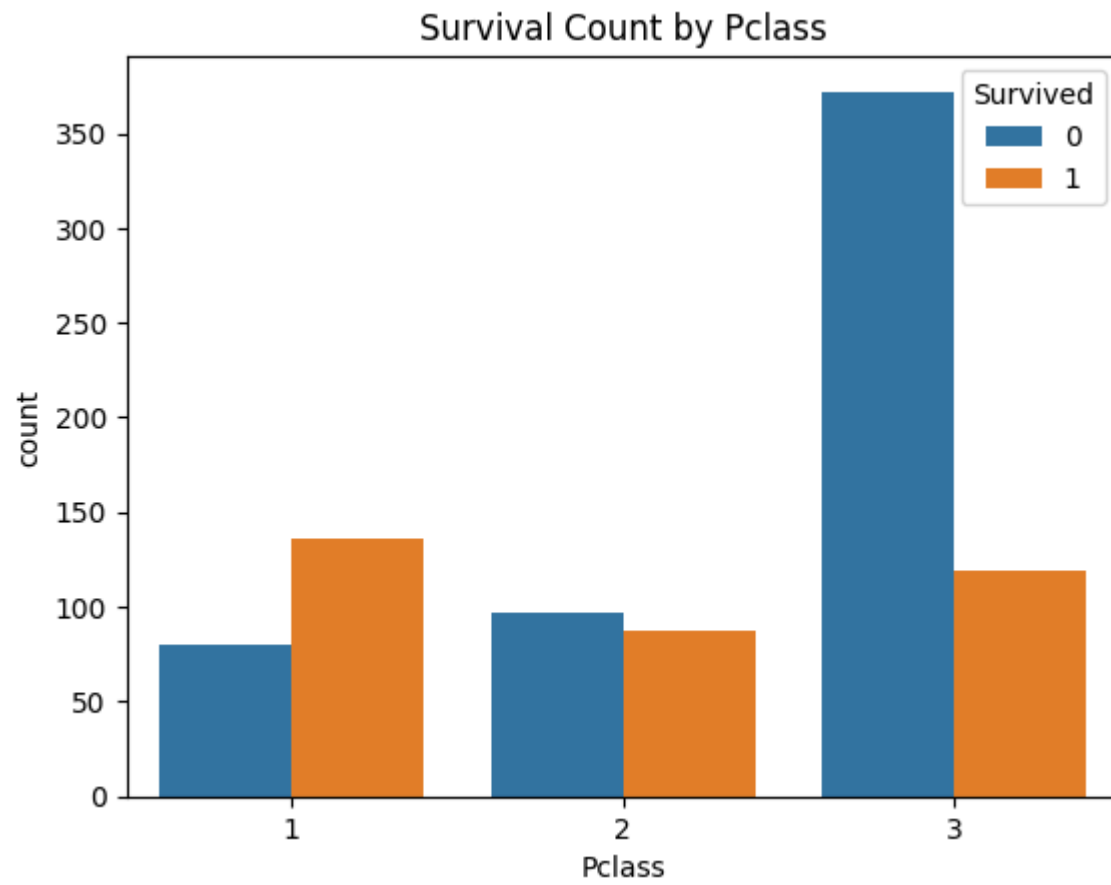


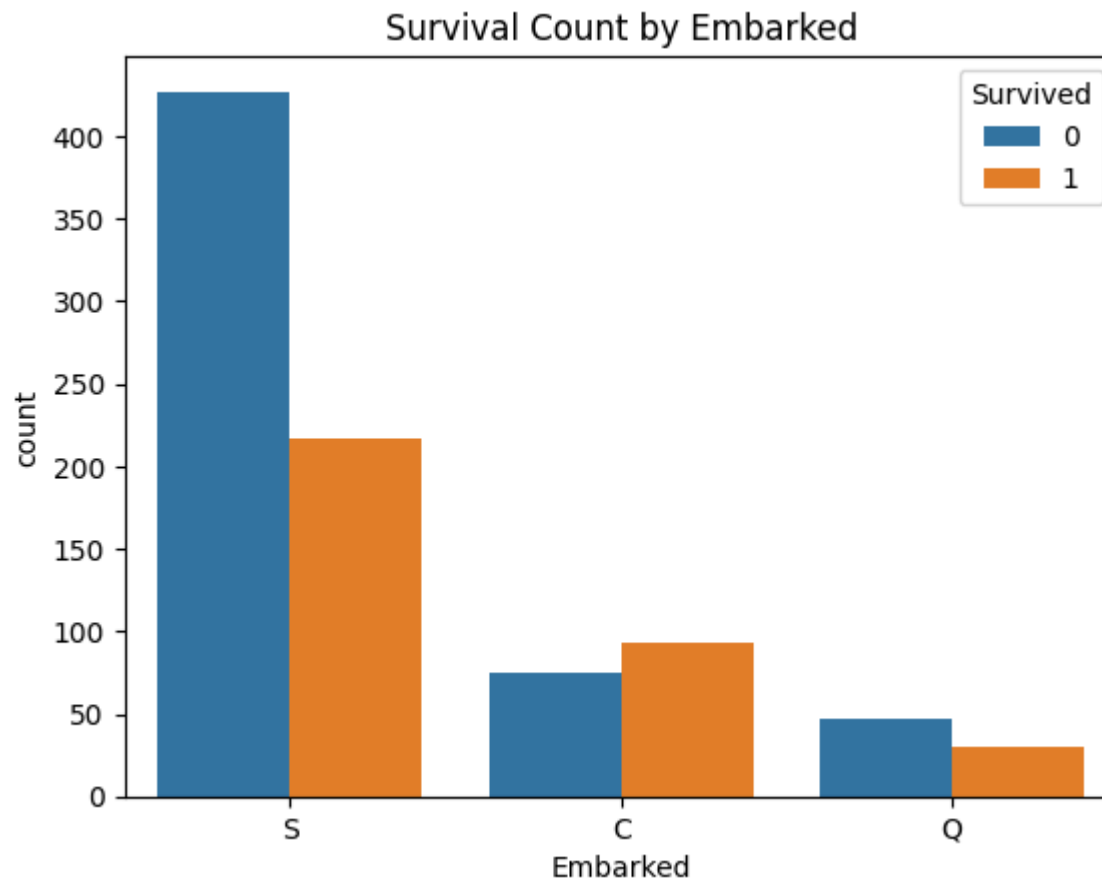
```
In [7]: # 5. Bivariate Analysis
sns.countplot(x='Sex', hue='Survived', data=train_df)
plt.title("Survival Count by Sex")
plt.show()

sns.countplot(x='Pclass', hue='Survived', data=train_df)
plt.title("Survival Count by Pclass")
plt.show()

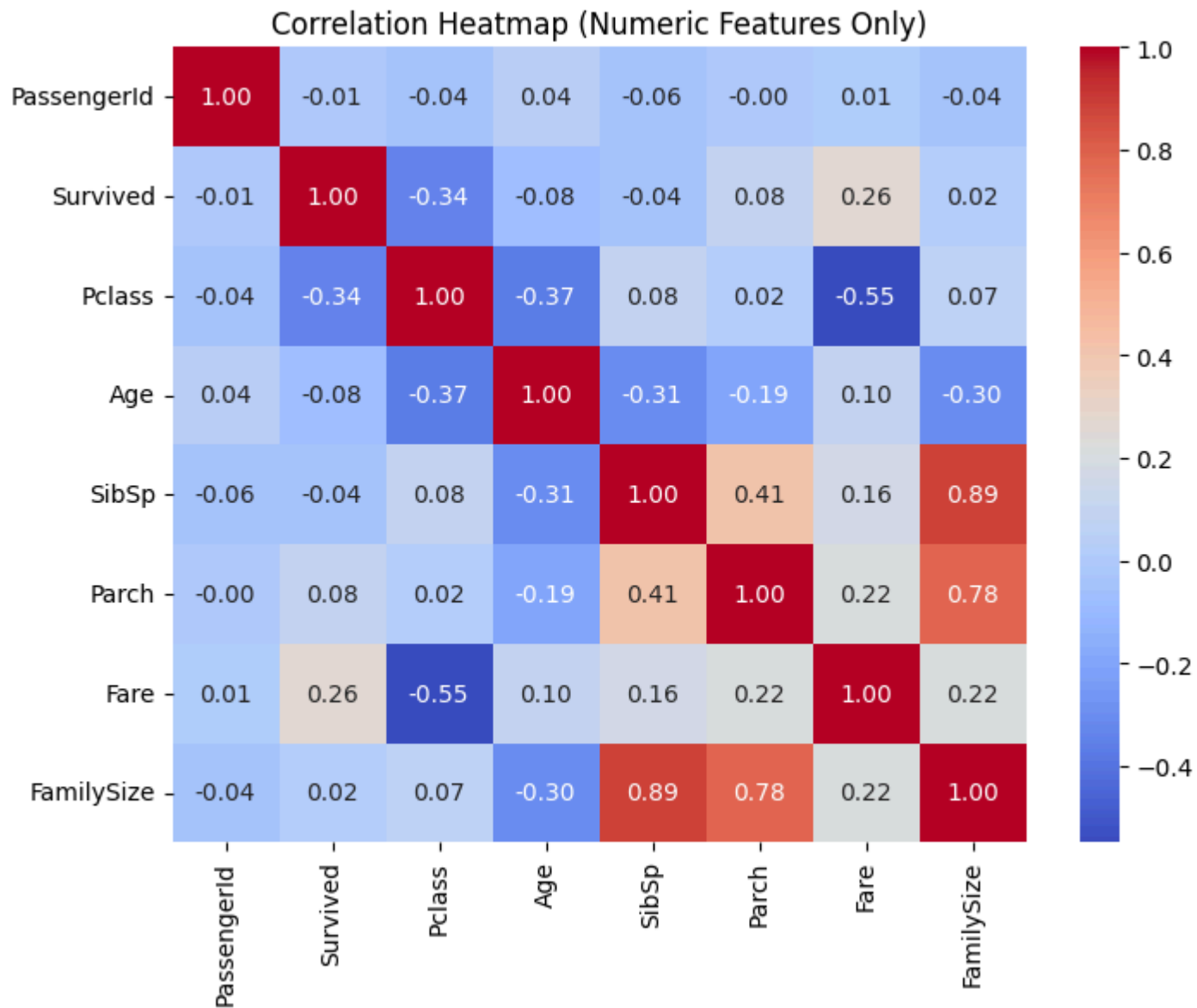
sns.countplot(x='Embarked', hue='Survived', data=train_df)
plt.title("Survival Count by Embarked")
plt.show()
```



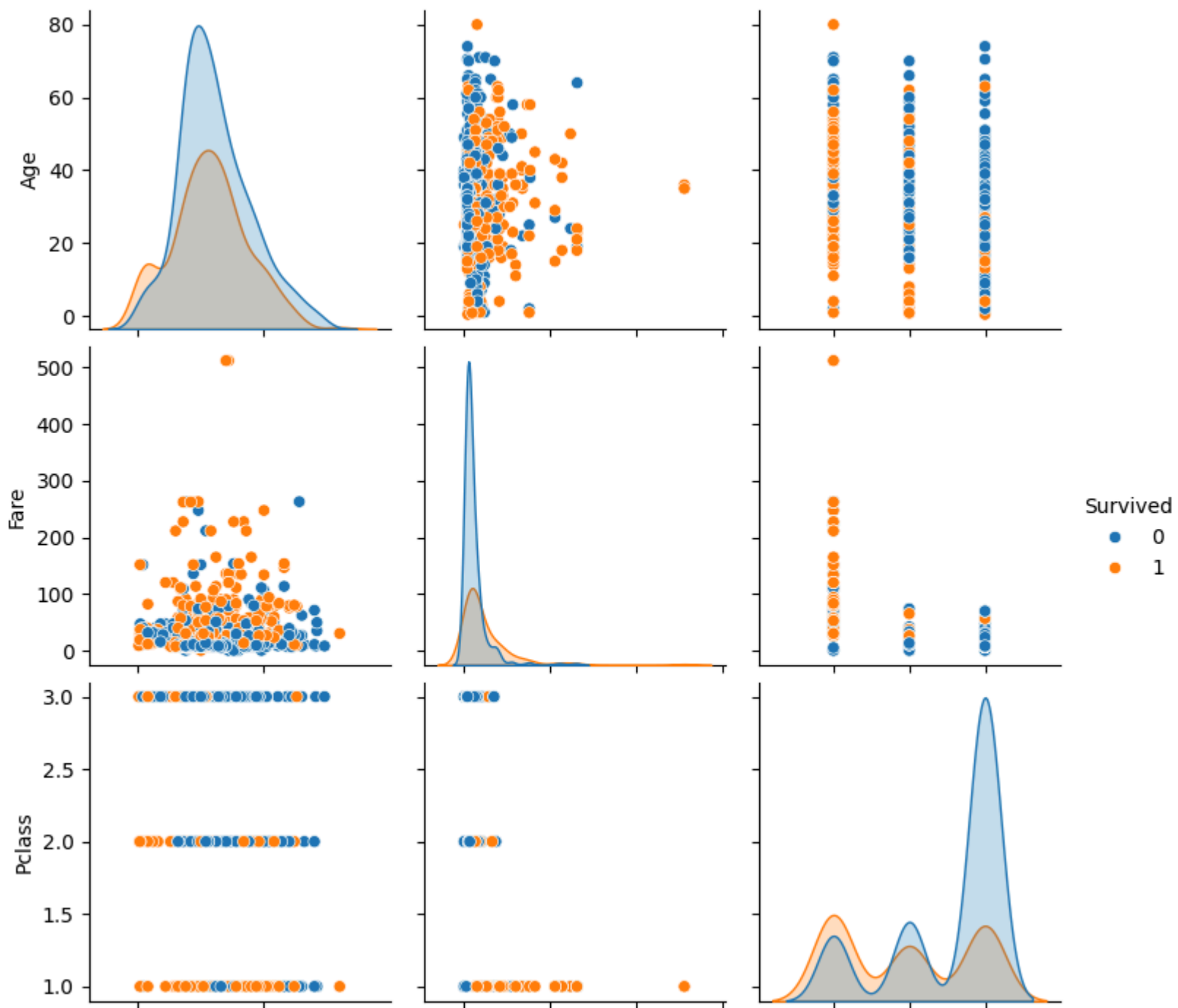




```
In [12]: # 6. Correlation Heatmap (fixed)
plt.figure(figsize=(8,6))
numeric_df = train_df.select_dtypes(include=['int64','float64']) # select only numeric columns
sns.heatmap(numeric_df.corr(), annot=True, cmap="coolwarm", fmt=".2f")
plt.title("Correlation Heatmap (Numeric Features Only)")
plt.show()
```



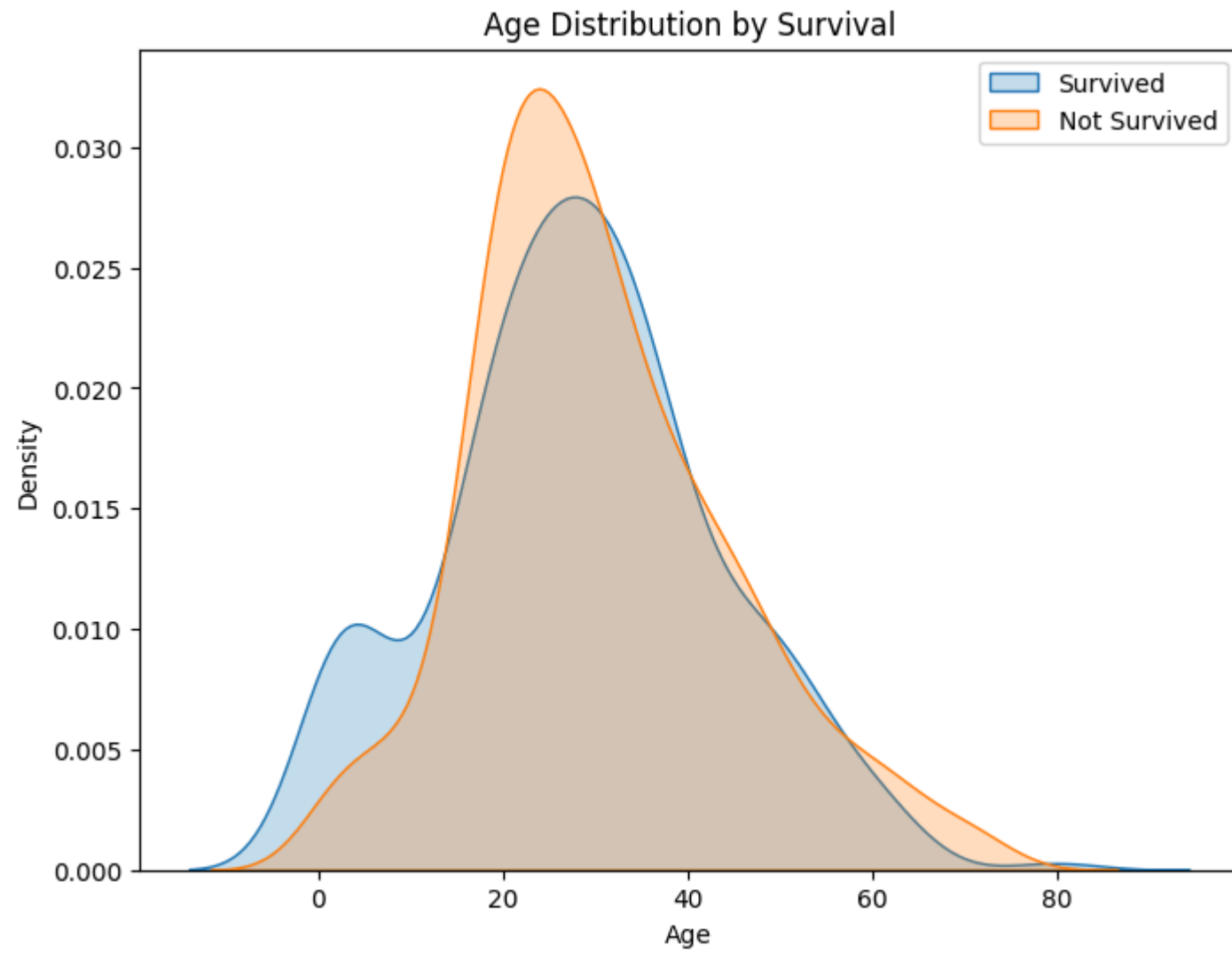
```
In [9]: # 7. Pairplot
sns.pairplot(train_df[['Survived', 'Age', 'Fare', 'Pclass']], hue='Survived')
plt.show()
```

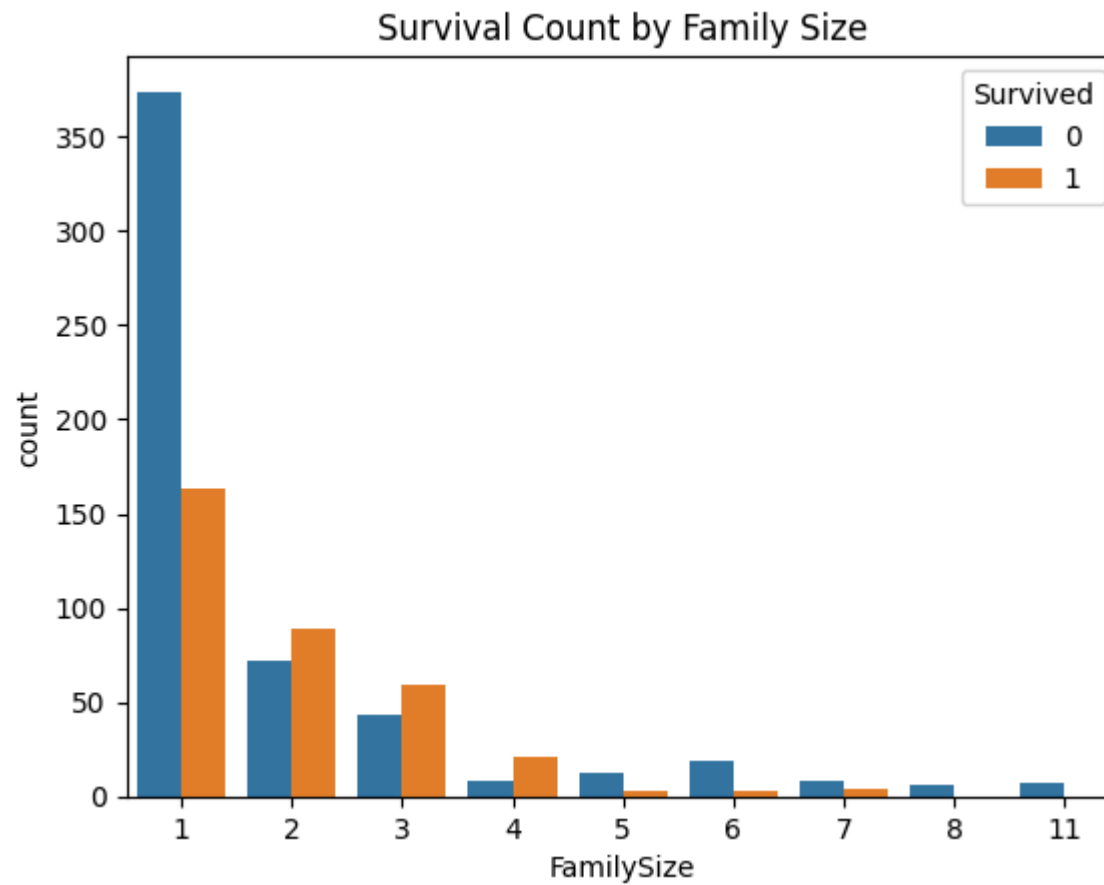


0 50 0 200 400 600 1 2 3
Age Fare Pclass

```
In [14]: # 8. Advanced Insights
plt.figure(figsize=(8,6))
sns.kdeplot(train_df.loc[train_df['Survived']==1, 'Age'], fill=True, label="Survived")
sns.kdeplot(train_df.loc[train_df['Survived']==0, 'Age'], fill=True, label="Not Survived")
plt.title("Age Distribution by Survival")
plt.legend()
plt.show()

train_df['FamilySize'] = train_df['SibSp'] + train_df['Parch'] + 1
sns.countplot(x='FamilySize', hue='Survived', data=train_df)
plt.title("Survival Count by Family Size")
plt.show()
```





```
In [15]: # 9. Summary of Findings
print("""
1. Majority of passengers did not survive (~62%).
2. Females had higher survival chances than males.
3. 1st class passengers had much better survival rates.
4. Passengers who paid higher fares were more likely to survive.
5. Many survivors were younger (children).
6. Larger families had lower survival rates compared to small families.
7. Missing values found in Age, Cabin, and Embarked columns.
""")
```

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In []: